

# CHEMISTRY MASTER SYLLABUS

CLASS 11 • NCERT / CBSE

## Chapter 1: Some Basic Concepts of Chemistry

- Matter
- Classification of Matter
- Physical and Chemical Properties
- Laws of Chemical Combination
- Atomic Mass
- Molecular Mass
- Mole Concept
- Stoichiometry
- Limiting Reagent
- Concentration Terms

## Chapter 2: Structure of Atom

- Atomic Models
- Electromagnetic Radiation
- Photoelectric Effect
- Atomic Spectrum
- Bohr Model
- Quantum Mechanical Model
- Dual Nature of Matter
- Heisenberg Uncertainty Principle
- Quantum Numbers
- Atomic Orbitals
- Electronic Configuration
- Aufbau Principle
- Pauli Exclusion Principle
- Hund's Rule

## Chapter 3: Classification of Elements and Periodicity in Properties

- Modern Periodic Law
- Modern Periodic Table
- Electronic Configuration
- Atomic Radius
- Ionic Radius
- Ionization Enthalpy
- Electron Gain Enthalpy
- Electronegativity
- Valency
- Periodic Trends

## Chapter 4: Chemical Bonding and Molecular Structure

- Chemical Bonding
- Octet Rule
- Lewis Structures
- Ionic Bond
- Covalent Bond
- Coordinate Bond
- Bond Parameters
- Valence Bond Theory
- Hybridization
- Molecular Orbital Theory
- Hydrogen Bonding
- VSEPR Theory

- Molecular Geometry

## Chapter 5: Thermodynamics

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- Thermodynamic System
- Surroundings
- State Functions
- Internal Energy
- Heat
- Work
- Enthalpy
- Heat Capacity
- Hess's Law
- Enthalpy Change
- Entropy
- Gibbs Energy
- Spontaneity

## Chapter 6: Equilibrium

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- Dynamic Equilibrium
- Law of Mass Action
- Equilibrium Constant
- Le Chatelier's Principle
- Chemical Equilibrium
- Ionic Equilibrium
- Acids
- Bases
- pH
- Buffer Solutions
- Solubility Product
- Common Ion Effect

## Chapter 7: Redox Reactions

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- Oxidation
- Reduction
- Oxidation Number
- Oxidising Agent
- Reducing Agent
- Balancing Redox Reactions
- Redox Titration
- Applications of Redox Reactions

## Chapter 8: Organic Chemistry – Some Basic Principles and Techniques

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- Organic Compounds
- Classification of Organic Compounds
- IUPAC Nomenclature
- Structural Isomerism
- Electronic Effects
- Reaction Intermediates
- Organic Reaction Mechanisms
- Purification Techniques
- Qualitative Analysis
- Quantitative Analysis

## Chapter 9: Hydrocarbons

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- Classification of Hydrocarbons
- Alkanes
- Alkenes
- Alkynes
- Aromatic Hydrocarbons
- Preparation of Hydrocarbons
- Physical Properties
- Chemical Properties
- Hydrocarbon Reactions
- Aromaticity